PROFINET as a Real-Time Industrial Ethernet for Automation Systems — Standard, Applications, Safety, Comparative Analysis, and Prototype

Rionela Kovaci
Electronic Engineering
Hochschule Hamm-Lippstadt
Lippstadt, Germany
rionela.kovaci@stud.hshl.de

Mohamed Abdo
Electronic Engineering
Hochschule Hamm-Lippstadt
Lippstadt, Germany
mohamed-sayed-mohamed.abdo@stud.hshl.de

Christian Percival
Electronic Engineering
Hochschule Hamm-Lippstadt
Lippstadt, Germany
christian.percival@stud.hshl.de

***Abstract*—PROFINET (Process Field Net) is the leading industrial Ethernet standard for real-time communication in automation systems, defined under IEC 61158 and IEC 61784. This paper provides a comprehensive technical analysis of PROFINET, covering its protocol architecture and the three communication classes; NRT, RT, and IRT which delivers cycle times from 100 ms down to 250 µs with sub-microsecond jitter. The functional safety extension PROFIsafe, standardized under IEC 61784-3-3, is analyzed for its "Black Channel" approach enabling up to SIL 3/PL e safety integrity across a broad range of industrial devices. Application areas spanning automotive manufacturing, process automation, food-and-beverage, and energy industries are examined. A systematic comparison of PROFINET against EtherNet/IP, POWERLINK, EtherCAT, Modbus ,and TCP is provided, evaluating cycle time, safety profile, topology, and market focus. Finally, a Raspberry Pi prototype implementing a Python-based PROFINET IO stack is presented, with device commissioning via PRONETA and protocol-level verification via Wireshark, demonstrating cyclic RT data exchange and validating the theoretical concepts in a laboratory setting.**

***Index Terms*—PROFINET, industrial Ethernet, real-time communication, PROFIsafe, IEC 61158, functional safety, Raspberry Pi, Wireshark, PRONETA, EtherCAT, EtherNet/IP, TSN**

I. Introduction

Modern industrial automation places increasingly stringent demands on communication networks. Production systems require the exchange of process data between PLCs, sensors, actuators, drives, and HMIs with latency guaran- tees, reliability, and in environments where functional safety is certified. Classical serial field buses like PROFIBUS, DeviceNet, and Modbus RTU were once used as the basic fabric of manufacturing communication for many years. However, their limited bandwidth, point-to-point wiring costs, and lack of integration with enterprise IT networks have driven the industry toward Ethernet-based alternatives [1], [2].

Standard IEEE 802.3 Ethernet, while offering high bandwidth and low cost, was designed for best-effort without timing guarantees. A 100 ms delivery delay is acceptable for file transfers but catastrophic for a robot arm awaiting a position setpoint, where one missed signal can cause a broken component or a safety hazard. Beyond latency, the TCP/IP stack introduces unpredictable jitter from retransmission, congestion control, and operating system scheduling. Factory floors require thousands of devices to synchronize with sub-millisecond precision, not merely to communicate [1].

PROFINET (Process Field Net) emerged from this transition as the open industrial Ethernet standard maintained by PROFIBUS & PROFINET International (PI), an organization with more than 1,700 member companies worldwide. PROFINET is standardized as IEC 61158 and IEC 61784 family of profiles and integrates the wide spread use of IEEE 802.3 Ethernet as physical and data link layer with real time extensions which were not present in office Ethernet. [3], [4] As of 2024, More than 35 million PROFINET devices have been deployed worldwide up to 2024, making it the largest industrial Ethernet standard in installed base. [5].

PROFINET solves the determinism problem through a layered approach: it bypasses the TCP/IP stack entirely for time-critical process data, injecting frames directly at Layer 2 using a proprietary EtherType, while retaining standard IP communication for configuration, diagnostics, and IT integration on the same physical cable. This dual-channel design allows a single network infrastructure to serve both real-time control and enterprise data collection. The key advantage over fieldbuses that required separate wiring for different data classes.

This paper is structured as follows. Section II describes the PROFINET standard, protocol architecture, and real-time communication classes. Section III surveys the main application domains. Section IV analyzes the PROFIsafe functional-safety profile. Section V provides a comparative evaluation against competing protocols. Section VI presents the Raspberry Pi prototype. Section VII offers conclusions and outlook.

II. The PROFINET Standard

A. Standardization and Governance

PROFINET is specified within the IEC 61158 series (fieldbus communication types) and IEC 61784-2 (real-time Ethernet communication profiles). Within IEC 61158, PROFINET corresponds to Type 10. The specification is organized across multiple sub-parts: application services are defined in IEC 61158-5-10, application protocols in IEC 61158-6-10, the communication profile in IEC 61784-2-3, a dedicated security profile in IEC 61784-6-3, cabling and installation in IEC 61784-5-3, and the PROFIsafe functional safety extension in IEC 61784-3-3 [5]. Governance is provided by the PI organization, which operates accredited test laboratories, a device certification program, and regional PI associations worldwide. A summary of the key standards is given in Table I.

TABLE I
KEY STANDARDS GOVERNING PROFINET

Standard	Scope
IEC 61158	Fieldbus specifications; PROFINET is Type 10
IEC 61784-2	RT and IRT performance requirements
IEC 61784-3-3	PROFIsafe functional safety profile (SIL 3/PL e)
IEC 61784-5-3	Cabling, connectors, and network infrastructure
IEC 61784-6-3	PROFINET security profile
IEEE 802.3	Ethernet physical and data-link layer base
ISO 13849	Functional safety basis (PL e / SIL 3)

B. Protocol Architecture

PROFINET inherits the IEEE 802.3 Ethernet frame at the physical and data-link layer but diverges from standard networking above that point. For time-critical process data, PROFINET bypasses the IP and TCP/UDP transport stack entirely, injecting frames directly above the data-link layer using EtherType 0x8892. These real-time frames carry cyclic IO data in the provider consumer model; an IO-Device produces output data at each cycle and the IO-Controller consumes it, and vice versa for input data [3], [4].

For non-time-critical operations initial configuration, parameter download, diagnostic reads, and remote procedure calls (RPC). PROFINET uses standard UDP/IP. This coexistence means a single cable and switch infrastructure can simultaneously carry 4 ms cycle RT control data and standard IP-based diagnostic traffic without interference, Which implements appropriate QoS priority queuing for EtherType 0x8892 frames.

The dominant application variant is **PROFINET IO**, which provides cyclic I/O data exchange between an IO-Controller (PLC, DCS) and IO-Devices (remote I/O blocks, drives, sensors, actuators). A second variant, PROFINET CBA (Component-Based Automation), supported modular plant design through DCOM-based component interconnection but was removed from IEC 61784-1 in its 4th edition (2014) and is no longer actively developed [6].

C. Real-Time Communication Classes

PROFINET defines three performance classes that address different latency and determinism requirements [4], [7].

1) Non-Real-Time (NRT): NRT uses the standard TCP/IP and UDP/IP stack for parameterization, configuration, and diagnostic data exchange. It supports IT integration services including OPC UA, SNMP, HTTP, and MQTT, enabling seamless connection of the PROFINET network to enterprise MES and ERP systems. No hard timing guarantees are provided; cycle times are in the order of 100 ms or more depending on network load.

2) Soft Real-Time (RT — Class A/B): RT communication bypasses the TCP/IP stack for process data, transmitting Layer 2 Ethernet frames identified by EtherType 0x8892 with an IEEE 802.1Q VLAN priority tag set to 6 for IO data and 5/7 for alarms. Standard managed switches supporting IEEE 802.1Q priority queuing are sufficient; no dedicated PROFINET switch ASIC is required. Achievable cycle times are 1–10 ms with jitter in the millisecond range, which is sufficient for digital I/O, valve control, frequency drives, and standard process automation tasks [3].

3) Isochronous Real-Time (IRT — Class C): IRT provides hardware-level synchronization using the Precision Transparent Clock Protocol (PTCP), an extension of IEEE 1588 PTP that compensates for propagation delay and switch forwarding delay. All devices and switches share a common distributed clock, and the send cycle is divided into a reserved IRT time slot and an open interval for RT and NRT traffic. Cycle times as low as 250 µs with jitter below 1 µs are achievable, meeting the requirements of coordinated multi-axis servo systems and high-speed CNC machining [1], [7]. IRT requires dedicated PROFINET-capable switches that implement the hardware scheduler; standard commercial off-the-shelf switches cannot be used for IRT.

D. Device Model and GSDML

PROFINET IO-Device is described by a GSDML (Generic Station Description Markup Language) file, an XML-based device description that specifies the slot and subslot structure, supported data lengths, module variants, and parameterization records. Engineering tools (Siemens TIA Portal, CODESYS, Rockwell Studio 5000) import the GSDML file to auto-configure the network project without manual parameter entry. At runtime, the IO-Controller discovers IO-Devices using LLDP (Link Layer Discovery Protocol) to build a topology map, and uses the Discovery and Configuration Protocol (DCP) to assign device names and IP addresses [3], [5].

E. Network Topology and Redundancy

PROFINET supports line, tree, star, and ring topologies. Many IO-Devices include integrated two-port switches with cut-through forwarding to support line topology without external switches, reducing wiring costs on the factory floor. Ring topology combined with the Media Redundancy Protocol (MRP) provides automatic reconfiguration within 200 ms (MRP) or 30 ms (MRPD for IRT networks) in the event of a link failure, meeting availability requirements in process and energy applications.

F. Evolution toward TSN

PROFINET Conformance Class D, introduced in version 2.4, integrates IEEE 802.1 Time-Sensitive Networking (TSN) capabilities. TSN provides hardware-independent time synchronization (IEEE 802.1AS), time-aware traffic shaping (IEEE 802.1Qbv), and frame preemption (IEEE 802.1Qbu) using standardized mechanisms rather than proprietary PROFINET extensions. This enables convergence of PROFINET control traffic with IT data on a single standards-based Ethernet fabric, The key enabler for Industry 4.0 and OT/IT integration architectures [5].

III. Application Areas

PROFINET is now used almost in every major industry. It combines real-time and isochronous communication classes, the PROFIsafe safety layer, and seamless IT integration through the NRT channel makes it suitable for a wider range of use cases than any single competing protocol.

A. Automotive Manufacturing

PROFINET is used the most in sector of car manufacturing. Many different machines working at the same time in a modern car assembly line.All of them need to stay in a sync with each other,like painting robots, welding robots, self driving vehicles, conveyor belts and cameras that check product quality. Complex multi-axis movements in body assembly, painting, and powertrain production happen with extremely precise timing,are made sure by PROFINET´s IRT mode . Leading car manufacturers have built their entire assembly lines around PROFINET, and by 2024 around 54% of automotive factories were already using PROFINET-based robotics [5].

B. Discrete Manufacturing and Robotics

PROFINET,outside of car factories, is heavily used in general machine building,electronics production, and consumer goods manufacturing to handle communication between PLCs, motor drives, and robots. Roughly 58% of industrial robots globally operated on PROFINET networks as of 2024 [5]. Integration with the OPC UA companion specification allows robot cells to publish structured process data to Manufacturing Execution Systems (MES) and ERP systems as part of Industry 4.0 data pipelines, with PROFINET serving as the real-time transport layer and OPC UA as the semantic data layer [7].

C. Process Automation

The automotive industry is the largest end-user market for PROFINET. IT includes field instruments, valve positioners, flow meters, and distributed control systems in the process industry, gases, and oil, water treatment, chemical plants, and pharmaceutical manufacturing. Acyclic read/write services of PROFINET IO enable a HART-over-PROFINET proxy device to remotely manage and configure HART field instruments without any interference with cyclic process control [5]. Instrument diagnostics, such as valve wear and sensor drift alerts, can be accomplished via the PROFINET cable used to carry the cyclic process value, which results in less wiring than parallel analog instruments.

D. Food, Beverage, and Pharmaceuticals

The hygienic manufacturing environment demands PROFINET components that are washdown-rated, meeting IP67/IP69K standards. In manufacturing and packaging lines, PROFINET network solutions link bar code readers, label printers, weigh scales, fill level sensors, and vision cameras into an integrated control system with track-and-trace capabilities that meet food safety regulations. In addition to PROFIsafe for safety control loops in GMP-compliant lines in the pharmaceutical industry, PROFNET's extensive alarm and diagnostic logging capability is crucial for FDA's 21 CFR Part 11 compliance [5].

E. Energy and Utilities

In power generation and grid protection applications such as electrical substations, wind turbine generators, and solar photovoltaic inverter parks, PROFINET serves as the communications protocol for interfacing with SCADA systems via communication between field devices. The use of an MRP ring is very important in such implementations, where it is mandatory to have a highly available network since any failure in the communication can result in the interruption of the power production process or grid protection process.

IV. PROFIsafe: Functional Safety over PROFINET

A. Background and Standardization

Industrial safety devices such as, emergency stop buttons, two-hand control units, light curtains, safety door switches, and overfill prevention sensors need to communicate with fail-safe controllers (F-CPUs) over networks not which were not originally designed for safety-critical operation. Any system failure that causes a safety function to fail when it should step in can result in injury or death.

PROFIsafe was developed specifically to solve this issue by adding a certified safety protocol layer above standard PROFINET. It was first released in 1998 for PROFIBUS DP, with Version 2 in 2005 extending support to PROFINET IO. PROFIsafe is standardized under IEC 61784-3-3 and has been independently certified by both TÜV SÜD and IFA (Institut für Arbeitsschutz, Germany) [5]. Millions of nodes are deployed globally, making it the market-leading functional safety protocol for discrete manufacturing and process automation.

B. Safety Integrity Levels

PROFIsafe supports functional safety communication up to:

- SIL 3 per IEC 61508 and IEC 62061 (probability of dangerous failure on demand below $10^{-9}\,\mathrm{h}^{-1}$);
- Performance Level e (PL e) per ISO 13849-1; and
- Category 4 per EN 954-1.

Meeting SIL 3 which is equivalent to having fewer than one undetected dangerous communication failure per 114,155 years of continuous use . A requirement that is satisfied by having multi-level error detection [5].

C. The Black Channel Principle

PROFIsafe uses the "Black Channel" concept that is defined in IEC 61784-3. The underlying communication medium of PROFINET IO and PROFIBUS, is treated as an untrusted and faulty channel. All safety assurances are implemented within the PROFIsafe layer itself, above the standard network stack, without relying on any safety properties of the network, hardware or cables. This architecture allows PROFIsafe to operate on any compliant PROFINET network without requiring safety-certified switches, cables, or connectors, which greatly reduces cost compared to architectures that require a separate safety bus [5]. The layered structure is shown in Fig. 1.

Safety Application (F-CPU $\leftrightarrow$ F-Device)
PROFIsafe Layer F-CRC (24/32-bit) + Consecutive No. + Watchdog + F-Address
PROFINET RT (Black Channel, untrusted)
IEEE 802.3 Ethernet Physical Layer (Cat 5e / Cat 6) (Black Channel, untrusted)

Fig. 1. PROFIsafe layered architecture using the Black Channel principle. Standard and safety IO data share the same network infrastructure.

D. Error Detection Mechanisms

The seven communication error classes that can occur in any Ethernet-based network are defined in IEC 61784-3:

- message repetition
- deletion
- insertion
- incorrect sequencing
- address corruption
- data corruption
- unacceptable dela

PROFIsafe counters all using four different mechanisms [5]:

F-CRC checksum: A 24 or 32-bit safety cyclic redundancy check is computed over the process data, consecutive number, and F-Address. A calculated string of numbers or characters is attached to a data packet, and when the device receives the message, it recalculates the value, if the result matches the sent checksum, then the data is verified as complete. Any single-bit or burst error in the data is detected with a residual error probability below 10^{-9} per message.

Consecutive number: A monotonically incrementing counter is embedded in every F-message. The receiver detects lost messages through gaps in the counter, or duplicated messages with repeated values, and incorrect ordering, when the values are not in sequence, and responds by a safe state.

Watchdog timer: The F-Device runs a watchdog, a timer, that is reset by each successfully received valid F-message. If no valid message arrives within the configured watchdog time (the F-WD-Time) , the timer will reach zero, and the F-Device autonomously switches its outputs to a safe state regardless of the state of the network or the F-Host.

F-Address: A source and destination address pair is embedded within the PROFIsafe protocol data unit. This prevents a correctly formed F-message intended for one F-Device from being accepted by a different F-Device that may have received it due to network miss-configuration or routing error. So F-messages intended for specific safety modules are not interpenetrated by the uncorresponding safety module with the a different F-address.

E. System Integration

Standard and safety IO data coexist in the same PROFINET network: non-safety IO-Devices are completely unaffected by the PROFIsafe layer, and safety F-Devices participate in both standard and safety communication simultaneously. F-Device certification requires successful completion of PROFIsafe conformance tests at PI-accredited laboratories, ensuring interoperability between F-Devices and F-Hosts from different manufacturers [5].

V. Comparison with Competing Protocols

Industrial Ethernet has a number of competing protocols, each with a particular architectural choice based on their respective tradeoffs in cycle time, topology, hardware costs, and marketing strategy. The most relevant competitors of PROFINET will be discussed below, and certain important characteristics are outlined in Table II [1], [7].

A. EtherNet/IP

EtherNet/IP is an Ethernet-based network designed by Rockwell Automation and supported by ODVA. It is the leading protocol in North American automated factories and relies on the Common Industrial Protocol (CIP) over plain TCP/IP for non-time-critical setup data as well as UDP/IP for cyclic data exchange. Unlike PROFINET, EtherNet/IP does not circumvent the TCP/IP layer and therefore cannot deliver more than a soft real-time performance, meaning approximately 1–10 ms cycle times, which is adequate for discrete IO and drives but too slow for high-end motion control systems. EtherNet/IP has the ability to run on unmodified standard commercial Ethernet hardware. CIP Safety supports SIL 3-rated functional safety on the same network connection [7].

B. EtherCAT

EtherCAT, developed by Beckhoff Automation and managed by the EtherCAT Technology Group (ETG), achieves the fastest cycle times among industrial Ethernet protocols, as low as 31.25 µs, through its "processing on the fly" scheme, where each slave reads and injects data directly into a passing Ethernet frame without buffering or re-transmission. However, it requires a custom-made ASIC per slave and enforces a master-slave bus or ring topology, complicating multi-PLC architectures. FSoE (Fail-Safe over EtherCAT) offers SIL 3 safety level capability.

C. Modbus TCP

Modbus TCP is a simple extension of the Modbus RTU protocol sent over a conventional TCP/IP connection and provides the highest level of industrial protocol interoperability. Nevertheless, this protocol is entirely non-deterministic, whereby 10–100 ms response time varies depending on the load and no real-time requirements can be provided. There are no official standards for adding functional safety features, which excludes Modbus TCP from any safety-critical applications [1].

D. POWERLINK

Ethernet POWERLINK was introduced by B&R and managed by the Ethernet POWERLINK Standardization Group (EPSG). It operates sub-millisecond cycle times on standard off-the-shelf Ethernet technology with a slot-communication network management (SCNM) approach, where each cycle is divided into two parts: the synchronous phase (only for isochronous data traffic), and asynchronous phase (non-isochronous data traffic). POWERLINK is specified in IEC 61784 and employs openSAFETY profile to ensure SIL 3 level functional safety. This protocol is extensively used in Europe by machine builders who prefer high-speed motion control without ASIC technology [7].

TABLE II
COMPARISON OF MAJOR INDUSTRIAL ETHERNET PROTOCOLS

Feature	PROFINET	EtherNet/IP	EtherCAT	Modbus TCP	POWER-LINK
Standard	IEC 61158	ODVA/IEC	ETG/IEC	Modicon	EPSG/IEC
Min. cycle	250 µs	1 ms	31 µs	10 ms	<1 ms
Safety	PROFIsafe SIL 3	CIP Safety	FSoE SIL 3	None	openSAFETY
Std. HW	RT: yes; IRT: no	Yes	No (ASIC)	Yes	Yes
Topology	Star/Ring/Line	Star/Ring	Line/Ring	Star	Line/Ring
TCP by-pass	Yes	No	Yes	No	Yes
Market	Europe/Global	N. Amer.	Motion	Legacy	Euro. OEMs

VI. RASPBERRY PI PROFINET PROTOTYPE

A. Objectives and Hardware Setup

A prototype was implemented on a Raspberry Pi 4 Model B (4 GB RAM, Gigabit Ethernet) running the standard Raspberry Pi OS to demonstrate PROFINET RT IO communication, device commissioning, and protocol-level analysis using accessible, low-cost hardware. The prototype operates at PROFINET Conformance Class A (RT), which does not require dedicated switch ASICs, and is therefore realizable entirely in software on a general-purpose Linux system.

The physical setup consists of the Raspberry Pi connected to a laptop which hosts the Siemens PRONETA commissioning tool and runs Wireshark for protocol capture.The focus of the prototype is protocol-layer correctness; demonstrating that the PROFINET DCP commissioning sequence, LLDP topology discovery, cyclic RT frame exchange, and frame timing can all be implemented and verified on a commodity platform.

B. Python-Based PROFINET Stack

The PROFINET IO protocol was implemented in Python on the Raspberry Pi, using raw Ethernet socket access (via the `socket` module with `AF_PACKET`) to send and receive Layer 2 frames directly without traversing the IP stack. The Python stack implements the following services required for a PROFINET IO-Device [4], [5]:

DCP (Discovery and Configuration Protocol): DCP operates at Layer 2 and is the mechanism by which the IO-Controller or commissioning tool discovers devices, reads their current name and IP address, and assigns a station name and IP address. The Raspberry Pi stack responds to DCP Identify All broadcast requests with a unicast DCP Identify Response containing the device type, vendor name, station name, and current IP configuration. It also handles DCP Set Name and DCP Set IP commands, storing the assigned values persistently. PRONETA uses these DCP services during the commissioning phase to configure the device before the PLC starts cyclic communication.

LLDP (Link Layer Discovery Protocol): The stack periodically transmits LLDP frames advertising the Raspberry Pi's chassis ID, port ID, time-to-live, and PROFINET-specific TLVs (Type-Length-Values) including the PROFINET port status. PRONETA reads these LLDP advertisements to build the physical topology map and verify that the device is connected to the correct switch port. Wireshark captures LLDP frames to confirm correct TLV structure.

Cyclic RT IO Data Exchange: After commissioning, the Python stack enters the cyclic data exchange phase. At each cycle, it constructs a PROFINET RT frame with EtherType 0x8892 containing: the PROFINET real-time header (FrameID in the range 0x8000–0xBFFF for RT Class 1), the IO Data block carrying the process output values, the IOCS (IO Consumer Status) byte, and the CycleCounter field incremented by 1 per cycle. The stack sends these frames at a configured interval of 4 ms and processes incoming frames from the IO-Controller that carry the corresponding output data.

C. Commissioning with PRONETA

Siemens PRONETA was used to perform network commissioning before starting cyclic communication. PRONETA scanned the subnet using DCP Identify All broadcasts and detected the Raspberry Pi as a PROFINET device, displaying its vendor name, device type, current IP address, and MAC address. The station name was then assigned via a DCP Set Name request, and DCP Set IP request to assign a static IP address within the controller's subnet. After commissioning, PRONETA's topology view populated from LLDP data correctly showed the Raspberry Pi connected to the switch port, confirming both DCP and LLDP were operating correctly.

D. Protocol Verification with Wireshark

All network traffic was captured using Wireshark with the PROFINET dissector plugin, which decodes Ether-Type 0x8892 frames and displays the PROFINET protocol fields in a human-readable tree view. This methodology is consistent with the approach used in related protocol analysis work [4]. The following observations were made from the captures:

DCP commissioning sequence: The capture confirmed the correct sequence of DCP messages during commissioning: a broadcast DCP Identify All request from PRONETA, followed by a unicast DCP Identify Response from the Raspberry Pi, then DCP Set Name and IP request/response pairs. All frames showed correct service IDs, service types, and option/sub-option fields as specified in the PROFINET standard [5].

RT frame structure: PROFINET RT cyclic frames were confirmed to carry the correct EtherType (0x8892), a FrameID in the RT Class 1 range, a correctly formatted CycleCounter field incrementing by 1 per cycle, and a DataStatus byte with the Primary bit set. The Wireshark PROFINET dissector decoded all fields without errors.

Cyclic timing: Using Wireshark's "IO Graphs" feature, the inter-frame arrival time of RT frames was plotted over a 30-second capture window. The measured average cycle period was 4.0 ms with a standard deviation of approximately 0.6 ms and occasional spikes up to 7–8 ms caused by Linux scheduler preemption under background CPU load. This is consistent with PROFINET Conformance Class A (RT) behavior, which tolerates jitter in the low-millisecond range and does not require hardware timestamping or a real-time kernel.

LLDP topology: LLDP frames from the Raspberry Pi were decoded by the Wireshark PROFINET dissector and showed correctly formatted PROFINET port status and chassis ID TLVs, consistent with PRONETA's topology display.

E. Limitations and Lessons Learned

The prototype demonstrates that PROFINET RT Class A communication including DCP commissioning, LLDP topology, cyclic Layer 2 IO data exchange, and GSDML-based auto-configuration can be fully implemented and verified in software on a commodity single-board computer. However, several fundamental limitations of general-purpose hardware for PROFINET were confirmed:

The standard Linux network stack introduces non-deterministic scheduling jitter that prevents cycle times below approximately 1 ms and produces occasional multi-millisecond spikes. Achieving PROFINET IRT (Class C, 250 µs, sub-microsecond jitter) requires dedicated hardware: PROFINET-certified switch ASICs with hardware schedulers (e.g., Hilscher netX series, Siemens ERTEC) and microcontroller hardware with PROFINET-specific real-time communication accelerators. These cannot be replicated in software regardless of operating system optimizations [7].

Despite these hardware constraints, the prototype provides a complete and verifiable demonstration of the PROFINET IO data model, the DCP commissioning workflow, and the RT frame structure offering a practical foundation for understanding the protocol concepts described throughout this paper.

VII. Conclusion

This paper presented a comprehensive technical overview of PROFINET as the leading real-time industrial Ethernet standard. The three-class communication model NRT, RT, and IRT delivers a flexible performance envelope covering the full range of automation requirements from non-critical configuration and IT integration through to isochronous multi-axis motion control with 250 µs cycle times and sub-microsecond jitter. The IEC 61158/IEC 61784 framework provides internationally standardized governance.

The PROFIsafe extension provides field-proven SIL 3/PL e functional safety communication over the same infrastructure used for standard IO, eliminating the need for separate safety wiring. Its Black Channel design places all safety assurance within the protocol layer, making it independent of the underlying network hardware and enabling safety signals to coexist on the same cable as standard automation data.

Against competing protocols, PROFINET offers the best balance of real-time performance, topology flexibility, IT integration, and ecosystem breadth. EtherCAT surpasses it in raw cycle-time performance but enforces a restrictive line topology and requires proprietary ASIC hardware in every slave. EtherNet/IP is the preferred choice in North American Rockwell-centric environments.

The Raspberry Pi prototype validated that PROFINET RT IO communication can be fully implemented and analyzed in software: the Python-based PROFINET stack correctly executed DCP commissioning via PRONETA, LLDP topology advertisement, and cyclic EtherType 0x8892 RT frame exchange at a 4 ms cycle interval. Wireshark protocol captures confirmed correct frame structure and measured cycle timing consistent with Class A behavior, while also illustrating the jitter limitations of general-purpose hardware that motivate dedicated ASIC solutions for IRT-class deployments.

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